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# Pharmacist-led telehealth disease management for patients with diabetes and depression

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**Abstract**

**Introduction:** The aim of this study was to determine whether a pharmacist-led telehealth disease management program is superior to usual care of nurse-led telehealth in improving diabetes medication adherence, haemoglobin A1C (A1C), and depression scores in patients with concomitant diabetes and depression.

**Methods:** Patients with diabetes and depression were randomized to pharmacist-led or nurse-led telehealth. Veterans with type 1 or type 2 diabetes, an A1C $\geq$ 7.5%, diagnosis of depression, and access to a landline phone were invited to participate. Patients were randomized to usual care of nurse-led telehealth or pharmacist-led telehealth. Patients were shown how to use the telehealth equipment by the nurse or pharmacist. In the pharmacist-led group, the patients received an in-depth medication review in addition to the instruction on the telehealth equipment.

**Results:** After six months, the pharmacist-led telehealth arm showed significant improvements for cardiovascular medication adherence (14.0; 95% confidence interval (CI) 0.4 to 27.6), antidepressant medication adherence (26.0; 95% CI 0.9 to 51.2), and overall medication adherence combined (13.9; 95% CI 6.6 to 21.2) from baseline to six-month follow-up. There was a significant difference in A1C between each group at the six-month follow-up in the nurse-led telehealth group (6.9 $\pm$ 0.9) as compared to the pharmacist-led telehealth group (8.8 $\pm$ 2.0). There was no significance in the change in patient health questionnaire-9 (PHQ-9) and Center for Epidemiologic Studies Depression Scale (CES-D) from baseline to follow-up in both groups.

**Discussion:** Pharmacist-led telehealth was efficacious in improving medication adherence for cardiovascular, antidepressants, and overall medications over a six-month period as compared to nurse-led telehealth. There was no significant improvement in overall depression scores.

**Keywords**

Telehealth, pharmacist, diabetes, depression, patient care management, eHealth

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## Introduction

The prevalence of concurrent major depression and significant depressive symptoms in patients with diabetes mellitus is estimated to be 11% and 31%, respectively – roughly two times the rate of the general population.[1] Their interaction has been recognized as particularly problematic as each condition negatively influences the other and results in decreased adherence to dietary recommendations and oral medications, more severe diabetes symptoms, poorer health related outcomes, additional complications, and increased healthcare costs.[2–5]

Medication adherence has been demonstrated to be worse among those with diabetes and depression.[2,6]

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Also, patients who are depressed are less likely to be adherent to individual appointments, manage lifestyle decisions, and may have little or no money to live a healthy lifestyle.[7] Depression was associated with increased symptoms associated with diabetes as well as less physical activity.[2] Multiple studies have shown an association between depression in diabetes and worse treatment adherence[8,9] and worse glucose control.[2,10] Poor adherence and worse disease control among those with diabetes and depression can be explained, in part, by the patient's low level of confidence in their ability to engage in self-care.[11] The association between depression and adherence in diabetes is important, as studies have shown that patients who are adherent to diabetes mellitus treatment have fewer hospitalizations.[12] Patients who are non-adherent to treatment have higher haemoglobin A1C (A1C), systolic and diastolic blood pressure, and low-density lipoprotein (LDL) cholesterol, all of which can cause hospitalizations and mortality.[12,13]

Pharmacist-led interventions have shown improvements in medication adherence and improvement in cardiovascular disease (CVD) risk factor controls.[9,14–16] Self-medication habits in patients with mental health conditions may be enhanced by improved health-provider communication.[17] One means of augmenting contact is through telehealth, which has been shown to reduce all-cause and diabetes-related hospitalizations.[18,19]

We hypothesized that a pharmacist-led telehealth disease management (pharmacist as care coordinator) program is superior to usual care (which is defined as nurse-led telehealth disease management –nurse as care coordinator) in improving the diabetes treatment adherence of patients with concomitant diabetes and depression, A1C, and depression scale improvement.

## Methods

### Research design

This pilot was an open-label randomized controlled study comparing the intervention arm of pharmacist-led telehealth disease management as an adjunct of usual care with the control arm of usual care of nurse-led telehealth disease management. Participants were assigned to the pharmacist-led telehealth arm or the usual care with nurse-led telehealth arm via a coin toss. The Institutional Review Board at the Providence VA Medical Center (RI, USA) approved the protocol, and all study procedures were conducted in accordance with the ethical standards of the Helsinki Declaration of 1975. Enrolment for this study began in May 2007 and ended in April 2011.

Using Veterans Health Information Systems and Technology Architectural (VISTA), we identified 210 patients with A1C $\geq$7.5% and diabetes diagnosis, of which 135 had a diagnosis of depression as well. A flyer was sent to the patients inviting them to participate, followed by a phone call to contact all eligible participants. Those patients who

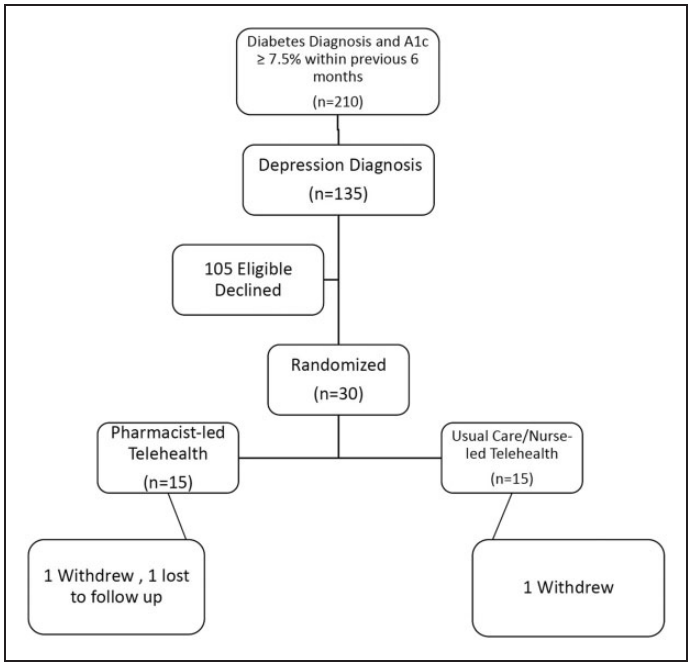


**Figure 1.** Study flow chart.

responded were screened and reviewed for inclusion and exclusion criteria. Only 30 agreed to participate and were randomized to intervention (pharmacist-led) or control (nurse-led) telehealth disease management (Figure 1).

### Population

Eligible patients were identified by a review of the Providence VA Medical Center's electronic medical record system. Veterans with type 1 or type 2 diabetes with an A1C $\geq$ 7.5% within the last six months, concomitant depression as defined by ICD-9 codes 311 or 296.2 to 296.3, and who were over 18 years old were eligible for the study. Patients must also have a landline phone and a three-prong electrical plug to operate the telehealth equipment.

Patients were excluded from the study if they were unable or unwilling to give consent, were without a qualifying A1C within the last six months, were pregnant (a pregnancy test would be administered to all women of child-bearing age), had a health literacy of less than an eighth-grade level, as measured by the rapid estimate of adult literacy in medicine (REALM) adult literacy assessment,[20] and if their major psychiatric, physical, or mental comorbidity precludes them from performing self-care or use of the telehealth technology. Patients that had only access to cell phones would not be able to use the telehealth equipment. Eligible patients were randomly assigned to usual care or telehealth through the flip of a coin.

### Telehealth Health Buddy®

The Health Buddy® appliance provided by Health Hero Network, Inc. used at the Providence VA Medical Center collects and transmits information about a patient's chronic condition, including vital signs, symptoms, and behaviours. The Health Buddy® is an electronic device with a monitor/screen that enables communication between providers and their patients. It is approximately the size of a seven-inch television with four easily visible blue buttons. Patients can be prompted to take required measurements with a variety of medical devices, including blood glucose monitors, scales, and blood pressure monitors. The Health Buddy® system automatically dials a toll-free number to send the information to a secure data collection centre. Although there are several different modules, our patients were "activated" into the diabetes module. At the time of the intervention, the depression module was not active. Each day, the Health Buddy® greeted the patient with a personalized greeting on the screen and asked them if they had been adherent to their medications and self-care issues. The Health Buddy® gave prompts to the patient to remind them to check their feet or take their medications, for example. After each session there were educational or trivia questions for the patient to answer before sending their results. The nurse or pharmacist received data from the Health Buddy® daily. Alerts of urgency and values outside of normal, such as out-of-range blood sugars, dietary intake, self-care behaviour, management of health-related symptoms, and mood symptoms were accessible via the secured website. In addition, patients were asked a number of different knowledge questions regarding diabetes, but those questions were only visible to the patient and were not assessed or evaluated. The nurse or pharmacist could contact the patient via telephone to discuss the care with the patient.

### Usual care visit

The usual care visit was with one of the two telehealth nurses. This one-on-one in-person visit lasted approximately 45 minutes and the patient was taught the basic functions and tasks of the telehealth technology, the Health Buddy® system, and how to operate the equipment. Via return demonstration, patients showed the nurse how to operate the Health Buddy®. They were encouraged to enter daily information that they obtained at home into the telehealth machine. The usual care visit also included a medication reconciliation performed at the visit by the telehealth nurse. When the nurse follows up with the patient for telehealth phone calls, they may work with the primary care physician (PCP) or nurse case manager assigned to the PCP to order lab work or initiate a care plan change. All subsequent interactions between the nurse and the patient were completed via telehealth equipment and phone contact.

### Telehealth pharmacist intervention visit

The pharmacist-led telehealth visit was a one-on-one in-person visit divided into two parts. There was only one pharmacist that was involved in the telehealth pharmacist portion of this study. The first part lasted approximately 45 minutes and replicated the usual care visit as described previously. For the second part of the visit, patients were asked to bring in all of their medications, including non-prescription and alternative medications. The telehealth research pharmacist answered questions and concerns about the medications, discussed side effects, how to take the medications, confirmed dosages, checked for omissions and appropriate indications, reviewed how to properly administer medications, and educated patients regarding their antidepressant and diabetes medications.

In addition, the telehealth research pharmacist educated the patient on the self-management of blood glucose and weight assessment. Similar to usual care, the pharmacist may make phone calls to the patient to discuss out-of-range results or to encourage the patient with positive reinforcement. All subsequent interactions between the pharmacist and the patient were completed via telehealth equipment and phone contacts. Pharmacists at the Providence VA Medical Center may have prescriptive authority to make changes to the patient's regimen or order lab tests, if necessary. Medication titrations were based on self-monitored glucose levels that had been validated and patient symptoms targeting improvement in A1C based on previously formatted medication titration algorithms as per national guidelines adapted to the existing Providence VA Medical Center formulary. The patient's PCP was alerted to all medication changes.

### Six-month follow-up appointment: intervention group

The six-month follow-up visit for patients in the intervention group was performed by the telehealth research pharmacist. Issues and concerns regarding treatment or medications were addressed. There was only a follow-up clinical appointment for the intervention group, not the usual care (nurse-led) group.

### Outcomes

The primary outcome was the comparison of the change in diabetes and depression medication adherence rate from baseline to six months between usual care and the intervention group. Adherence was measured with pill counts and Grymonpre adherence formula.[21] The Grymonpre adherence formula utilizes pill counts to calculate an adherence percentage. It was only used on regular-use medications for diabetes, CVD, and depression. Secondary outcomes were the change in A1C value and change in patient health questionnaire-9 (PHQ-9) and Center for Epidemiologic Studies Depression Scale (CES-D) depression score from baseline to six months.[22] The PHQ-9 and CES-D are validated measures of depressive symptom burden for primary care patients, with higher scores indicating a greater depressive burden.[22,23]

### Data collection

Demographic variables (i.e. age, sex) and comorbid conditions (i.e. coronary artery disease) were obtained from the medical record at baseline and confirmed by interview. Physical assessment values of blood pressure (i.e. systolic and diastolic blood pressure) and weight were obtained at baseline and at the six-month follow-up visit. Laboratory values of A1C and LDL cholesterol were obtained at baseline and six-month follow-up visit. Medication adherence was assessed via pill counts and Grymonpre formula at baseline and at six-month follow-up visit. All diabetes, cardiovascular, and antidepressant medications were counted and plugged into the Grymonpre formula that used quantity dispensed subtracted by quantity remaining, divided by the number of prescribed tablets per day.[21] Depression scores were assessed using the PHQ-9 and CES-D questionnaires[22,23] at baseline and at six-month follow-up. The PHQ-9 depression questionnaire is a nine-question Likert scale assessment of depression symptoms and is scored out of 27 points. A lower score indicates less symptoms and a score of 10 or higher indicates moderate or severe symptoms of depression. The CES-D questionnaire is a 20-question Likert scale used to assess depression in research and clinical settings in wide age ranges. A score of 16 or higher indicates a clinically significant risk of depression.

### Statistical analysis

Baseline characteristics were compared between nurse-led telehealth and pharmacist-led telehealth groups utilizing *t*-tests for continuous variables and chi-square tests for discrete variables. To compare the differences in medication adherence, we used a *t*-test. We used the chi-square test to evaluate achievement in A1C goals, and *t*-tests for change in depression scores and A1C. All statistical analyses were conducted using SPSS 22, (IBM Corp., released 2013, IBM SPSS Statistics for Mac, version 22.0, Armonk, NY, USA). Statistical significance was defined as $p < 0.05$. Data are presented as mean $\pm$ standard deviation and 95% confidence intervals (CIs).

## Results

Approximately one in five eligible patients who were approached agreed to participate. Of the 135 eligible patients, 30 patients agreed to participate and were randomized to pharmacist-led telehealth or nurse-led telehealth. One patient withdrew from each group and one additional patient was lost to follow-up. Overall, 27 patients were included in the final analysis, 13 in the pharmacist-led telehealth group, and 14 in the nurse-led telehealth group (Figure 1).

Compared to the control arm of nurse-led telehealth, pharmacist-led telehealth had lower baseline levels of systolic blood pressure ($122.8 \pm 18.6$ mmHg vs. $137.4 \pm 18.4$ mmHg) and diastolic blood pressure ($69.5 \pm 7.8$ mmHg vs. $75.8 \pm 8.1$ mmHg). In addition, the pharmacist-led group had a lower baseline medication adherence score ($31.1 \pm 24.6$) compared to usual

**Table 1.** Baseline characteristics.

| Characteristic | Cases/pharmacist-led telehealth ($n = 15$) | Controls/nurse-led telehealth ($n = 15$) | $p$-value |
|---|---|---|---|
| Age | 63.1 ± 12.2 | 60.5 ± 9.4 | 0.53 |
| Men (%) | 100 | 86.7 | 0.48 |
| Weight (kg) | 102.4 ± 17.9 | 105.7 ± 17.7 | 0.61 |
| Systolic blood pressure (mm Hg) | 122.8 ± 18.6 | 137.4 ± 18.4 | 0.04[a] |
| Diastolic blood pressure (mm Hg) | 69.5 ± 7.8 | 75.8 ± 8.1 | 0.04[a] |
| A1C (%) | 8.9 ± 2.1 | 7.6 ± 1.1 | 0.05 |
| Total cholesterol (mg/dl) | 176.6 ± 43.8 | 161.9 ± 28.6 | 0.31 |
| HDL cholesterol (mg/dl) | 36.1 ± 9.1 | 35.2 ± 9.1 | 0.80 |
| Triglycerides (mg/dl) | 242.4 ± 128.1 | 234.2 ± 138.7 | 0.88 |
| LDL cholesterol (mg/dl) | 103.1 ± 39.7 | 87.0 ± 24.8 | 0.22 |
| Smoker (%) | | | |
| Never | 6.7 | 13.3 | 0.59 |
| Current | 26.7 | 13.3 | |
| Former | 66.7 | 73.3 | |
| Highest level of education: (%) | | | |
| <High school | 13.3 | 13.3 | 0.80 |
| High school | 13.3 | 26.7 | |
| Some college | 40.0 | 40.0 | |
| Associates degree | 20.0 | 6.7 | |
| Bachelor's degree | 13.3 | 13.3 | |
| Advanced degree | 0 | 0 | |
| Age diabetes diagnosed (in years) | 50.3 ± 13.4 | 46.1 ± 12.3 | 0.40 |
| Coronary artery disease (%) | 66.7 | 33.3 | 0.14 |
| Hypertension (%) | 80.0 | 66.7 | 0.68 |
| Hyperlipidaemia (%) | 93.3 | 80.0 | 0.60 |
| Heart failure (%) | 26.7 | 6.7 | 0.33 |
| COPD (%) | 13.3 | 6.7 | 1.00 |
| Anxiety (%) | 26.7 | 26.7 | 1.00 |
| Depression (%) | 100 | 100 | 1.00 |
| PHQ-9 ($n = 13$) | 13.2 ± 7.1 | 7.3 ± 4.1 | 0.80 |
| CES-D ($n = 15$) | 30.4 ± 12.0 | 21.9 ± 6.7 | 0.12 |
| Grymonpre ACE inhibitor baseline | 35.5 ± 30.1 | 55.3 ± 35.8 | 0.27 |
| Grymonpre insulin long-acting baseline | 29.2 ± 21.6 | 72.5 ± 20.0 | 0.01 |
| Grymonpre insulin short-acting baseline | 42.4 ± 10.8 | 9.9 ± 14.0 | 0.12 |
| Grymonpre insulin combined baseline | 31.1 ± 24.6 | 73.6 ± 18.5 | 0.005[a] |
| Grymonpre statin baseline | 28.5 ± 32.6 | 51.1 ± 23.5 | 0.13 |
| Grymonpre antidepressant baseline | 25.1 ± 29.8 | 51.3 ± 36.9 | 0.08 |

[a] $p < 0.05$.
ACE: angiotensin converting enzyme; A1C: glycated hemoglobin; COPD: chronic obstructive pulmonary disease; HDL: high density lipoprotein; LDL: low density lipoprotein.

care, which had a baseline of 73.6 ± 18.5. Baseline characteristics were similar in other characteristics including the depression questionnaires (PHQ-9 and CES-D; Table 1).

After six months, the pharmacist-led telehealth arm showed significant improvements in cardiovascular medication adherence with the Grymonpre formula (14.0; 95% CI 0.4 to 27.6), antidepressant medication adherence (26.0; 95% CI 0.9 to 51.2), and all medication adherence combined (13.9; 95% CI 6.6 to 21.2) from baseline to six-month follow-up, whereas the usual care arm had no significant improvement from baseline to six-month follow-up for any of the medication adherence categories (cardiovascular, diabetes and insulin, antidepressants, adjunct antidepressants, and all medications combined). There were no significant differences from baseline to follow-up in the medication adherence for cardiovascular medications (10.3; 95% CI –10.0 to 30.6), diabetes medications and insulin (8.9; 95% CI –20.0 to 37.4), antidepressants (8.5; 95% CI –54.4 to 37.4), adjunct antidepressants (15.3; 95% CI –40.4 to 71.1), and all medications combined

**Table 2.** Medication adherence – Grymonpre adherence percentage.[1]

| Variable | Pharmacist-led: mean baseline adherence | Nurse-led: mean baseline adherence | Baseline mean difference between the groups 95% CI | Pharmacist-led: mean six-month adherence | Nurse-led: mean six-month adherence | Follow-up mean difference between the groups 95% CI | Pharmacist baseline to follow-up Mean difference 95% CI | Nurse baseline to follow-up Mean difference 95% CI |
|---|---|---|---|---|---|---|---|---|
| Cardiovascular medications | 18.9 ± 19.8 ($n = 15$) | 25.5 ± 22.5 ($n = 14$) | 15.1 (–6.4 to 36.6) p=0.16 | 51.2 ± 25.8 ($n = 12$) | 53.8 ± 19.0 ($n = 11$) | 10.3 (–10.0 to 30.6) | 14.0 (0.4 to 27.6)[a] | 5.9 (–10.1 to 21.8) |
| Diabetes medications and insulin | 35.5 ± 29.7 ($n = 14$) | 46.9 ± 24.9 ($n = 12$) | 11.3 (–11.0 to 33.7) | 42.2 ± 41.6 ($n = 13$) | 51.1 ± 25.8 ($n = 12$) | 8.9 (–20.0 to 37.4) | 3.9 (–12.7 to 20.6) | 2.6 (–15.4 to 20.6) |
| Antidepressants | 22.6 ± 23.2 ($n = 12$) | 54.2 ± 33.4 ($n = 10$) | 26.2 (–3.4 to 55.9) | 64.4 ± 34.1 ($n = 6$) | 54.0 ± 40.0 ($n = 7$) | 8.5 (–54.4 to 37.4) | 26.0 (0.9 to 51.2)[a] | 15.3 (–24.6 to 55.2) |
| Adjunct antidepressants | 34.5 ± 38.1 ($n = 4$) | 34.8 ± 25.1 ($n = 4$) | 7.4 (–31.6 to 46.4) | 50.5 ± 40.6 ($n = 4$) | 65.5 ± 34.3 ($n = 4$) | 15.3 (–40.4 to 71.1) | 42.0 (–415.4 to 499.4) | 7.7 (–59.6 to 74.9) |
| All medications combined | 38.0 ± 19.5 | 50.7 ± 22.3 | 12.7 (–3.2 to 28.6) | 54.5 ± 22.5 | 56.7 ± 15.2 | 2.3 (–13.7 to 18.3) | 13.9 (6.6 to 21.2)[a] | 4.6 (–14.0 to 23.1) |

[a]Significance ($p < 0.05$).
CI: confidence interval.

**Table 3.** Haemoglobin A1C change and change in depression scores.

| Variable | Nurse-led baseline | Pharmacist-led baseline | *P*-value | Nurse-led follow-up | Pharmacist-led follow-up | *P*-value | Mean change from baseline nurse-led (95% confidence interval) | Mean change from baseline pharmacist-led (95% confidence interval) |
|---|---|---|---|---|---|---|---|---|
| Haemoglobin A1C (A1C) | $7.6 \pm 1.1$ | $8.9 \pm 2.1$ | 0.05 | $6.9 \pm 0.9$ | $8.8 \pm 2.0$ | 0.006[a] | –0.42 (–0.21 to 1.04) | 0.40 (–1.01 to 0.23) |
| PHQ-9 | $7.3 \pm 4.1$ | $13.2 \pm 7.1$ | 0.80 | $7.7 \pm 3.7$ | $10.2 \pm 4.5$ | 0.33 | 0.67 (–2.95 to 4.28) | –0.67 (–7.19 to 8.69) |
| CES-D | $21.9 \pm 6.7$ | $30.4 \pm 12.0$ | 0.12 | $23.8 \pm 6.7$ | $27.4 \pm 12.5$ | 0.60 | 3.50 (–0.71 to 7.71) | –2.83 (–13.42 to 7.75) |

[a]Significance ($p < 0.05$).

(2.3; 95% CI –13.7 to 18.3) in pharmacist-led telehealth versus usual care (Table 2).

There was no significant difference between pharmacist-led and nurse-led telehealth from baseline to follow-up in each group. Although there was no significance in the change in A1C value from baseline to follow-up in both nurse-led and pharmacist-led telehealth, there was a significant difference in A1C between each group at the six-month follow-up, with nurse-led telehealth $6.9 \pm 0.9$ as compared to pharmacist-led telehealth $8.8 \pm 2.0$. In addition, there were no statistically significant changes with either PHQ-9 or CES-D depression screening tools from baseline to six-month follow-up or the mean change from within each group (Table 3).

## Discussion

Pharmacist-led telehealth achieved a greater attainment of medication adherence for cardiovascular, antidepressant, and overall medication adherence. Although both nurse-led and pharmacist-led telehealth reviewed the patient's medications, the pharmacist-led telehealth arm spent time reviewing the medication regimen for omissions, inappropriate dosing, and how to properly administer their medications. The pharmacist-led and nurse-led telehealth resulted in fairly identical outcomes for change in A1C, change in adherence score, and depression scores over the six-month study. The nurse-led group achieved A1C change, which was more significant from baseline to follow-up, despite improvements in adherence in the pharmacist-led group. This may have been due to more extensive experience in utilizing telehealth in the nurse-led group as compared to the pharmacist-led group. Another explanation is that this may have been due to changes that the primary care providers made through nurse-led disease management as compared to the pharmacist-led arm of the study, which may have been more cautious or tentative with medication changes due to less familiarity with the patients. The depression scores were higher in the pharmacist-led group in both PHQ-9 and CES-D questionnaires, but were not significant for either, due to the low numbers of patients in each group. Increased depression may have led to less significant impact on medication adherence and other self-care skills. In addition, a follow-up of six months may not have shown the full benefit of the telehealth and telecare interventions.

There are several potential explanations for the overall outcomes. The co-occurrence of diabetes and depression is associated with poorer physical and mental functioning, thus limiting self-care, the basis of diabetes management.[8] There is a large pill burden for patients with diabetes and depression, and it has become increasingly complex to manage patients with diabetes[24] due to the increased number of diabetes medications and novel delivery systems. Patients enrolled in this study had approximately nine medications for cardiovascular, diabetes, and mental health conditions. This number did not include medications for other chronic illnesses such as glaucoma or a dermatologic condition. Patients with multiple co-morbid conditions, especially depression, are more likely to receive substandard preventive diabetes care because of the competing demands for primary care provider's (PCP) time.[4] These problems may be compounded by a lack of referral of patients with depression and diabetes to disease management programs due to provider beliefs that the patient will not be able to self-manage their diabetes because of their mental health condition.[25] Therefore, patients that need disease management services the most may not be offered this option.

A study of telehealth with chronic illnesses who had a history of inpatient admissions for their conditions were found to have more emergency admissions and/or death as compared to control patients with no telehealth.[26] Varied intervention options may help to improve individual patient outcomes. This study shows that within six months, pharmacist-led telehealth has similar outcomes to nurse-led telehealth. Patients with co-occurrence of depression and diabetes are susceptible to poorer outcomes, and methods that may assist them will be important.[27–29] The daily monitoring of diabetes control and depressive symptoms

through the use of telehealth may be an alternative method to help patients manage their condition.

Our study does have some limitations. One limitation was that only one in five participants responded to the inquiry of the study and were willing to participate. In addition, patients enrolled in telehealth must also have a landline phone and a three-prong electrical plug to operate the telehealth equipment. Many patients had started to use cell phones and other phone services that were not compatible with the Health Buddy® and, therefore, were ineligible for enrolment. Another limitation of our study is the usual care group. Initially, we were comparing pharmacist-led telehealth to patients not enrolled in telehealth. During the study's inception, the Department of Veterans Affairs (VA) policy of eligibility for telehealth was interpreted for appropriateness of use. It was interpreted that if a patient qualified for telehealth, they should be offered telehealth and not delayed in treatment. Waiting six months to be placed on telehealth was not acceptable; therefore, we compared pharmacist-led telehealth to nurse-led telehealth for all eligible patients. In addition, another limitation of the study was in the change of the depression scoring system used, which was changed midway through the study. The study initially used PHQ-9, but was modified and changed to the CES-D scale to capture depression changes more easily. This resulted in less comparable depression scores in each group.

In conclusion, pharmacist-led telehealth was efficacious in improving medication adherence for cardiovascular, antidepressants, and overall medications over a six-month period as compared to nurse-led telehealth. This study demonstrates that pharmacist-led telehealth is not superior to nurse-led telehealth for patients with diabetes and concomitant depression or in their change in A1C or depression scale scores.

## Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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